Task-Aware Object Selection Using FPGA-Based Two-Stage Approximation

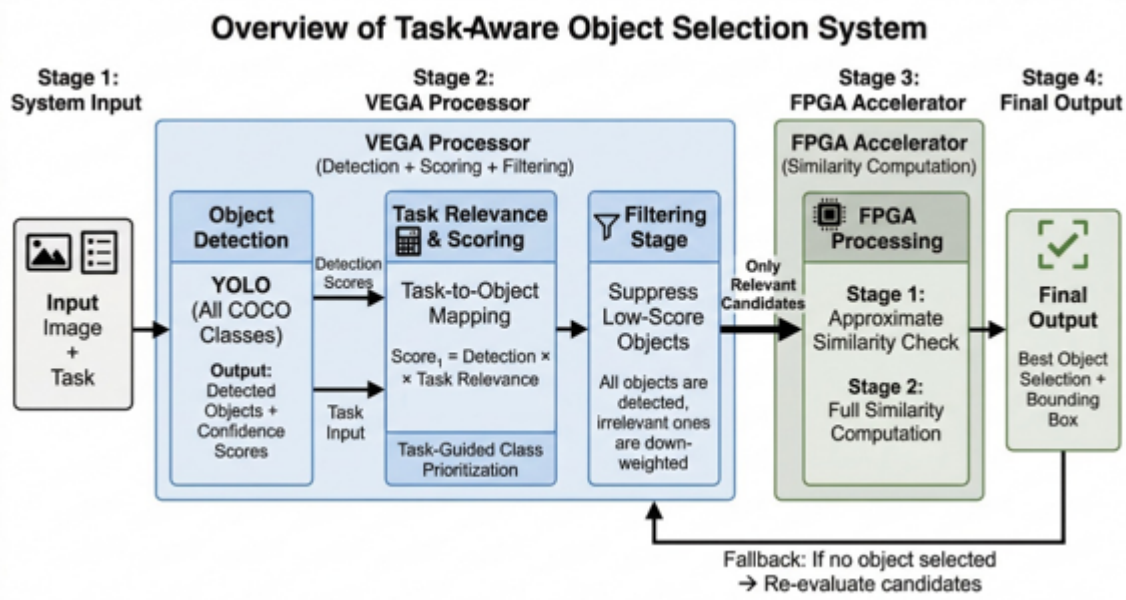
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I. OVERVIEW

ChipUp presents a hardware-accelerated **Task-Aware Object Selection Engine**, designed and integrated into the VEGA RISC-V SoC for the DVCon India 2026 Design Contest.

The system identifies the object most relevant to a given task in a scene: YOLOv8 detects candidate objects and assigns each a confidence score, which is combined with a task-relevance weight to rank the top five candidates. These five confidence/weight pairs are handed off to our custom hardware accelerator for final decision-making.

Our custom accelerator — implemented as a custom **lightweight** AXI4-integrated IP block which is within the VEGA SoC — computes a weighted score for each of the five candidates directly in hardware, and selects the best (primary) and second-best (alternative) match with deterministic, low-latency execution. It replaces the organiser's example accelerator in the SoC, communicating with the VEGA RISC-V core over the same AXI4 register interface used throughout the platform.



II. DESIGN HIGHLIGHTS

Novelty

- Hardware/software co-design: detection + scoring in software(YOLOv8), final decision-making in a dedicated AXI4 hardware engine.

Accuracy

- **5/5 test cases** verified bit-accurate against hand-calculated expected scores.
- Covers every decision path: ***high-priority match***, ***medium-priority fallback***, ***no-high-tier case***, ***independent data point***, ***all-filtered edge case***.

Latency

- Full 5-test suite completes in under 500 µs (simulated).
- Fixed-cycle decision logic — deterministic, no variable-latency stalls.

Resource Footprint

- **+0.41% LUTs, +0.20% registers, +0.60% DSP** over the baseline VEGA SoC.
- Coexists comfortably with CPU, DDR3, Ethernet, UART — no resource pressure on the Genesys-2 Kintex-7.

Power

- **0.003 W** — just **~0.5%** of total SoC power (0.568 W).
- Negligible overhead, ideal for power-constrained edge deployment.

III. ACCELERATOR ARCHITECTURE

i) Accelerator_Top.v

The top-level wrapper: this instantiates the AXI4 slave interface, the AXI4 master interface, and the scoring core **(top2_comparator**), and wires them together. This is the module instantiated directly inside **Top.vhd**, replacing the provided dummy accelerator.

ii) axi_slave_interface.v

Implements the AXI4-Lite slave register interface through which the RISC-V core (VEGA/AS1061) configures the accelerator and reads back results. Register map:

Offset	Name	R/W	Description
0x00 – 0x20	CONF_WEIGHT_0..4	W	5 pairs: {weight[31:16], confidence[15:0]}
0x28	START	W	Bit 0: pulse to begin computation
0x30	RESULT_0	R	{primary_score, primary_index}
0x38	RESULT_1	R	{alternative_score, alternative_index}
0x40	VALID_MASK	R	Bitmask of candidates above threshold
0x48	READY	R	Legacy busy-flag (kept for compatibility)
0x50	RESULT_BASE_ADDR	W	Memory address for AXI write-back of results
0x58	STICKY_DONE	R	Latched completion flag — set only once the AXI master finishes writing results to memory, cleared on the next START. Used by firmware instead of READY to avoid a race condition between back-to-back test runs.

iii) axi_master_interface.v

Implements the AXI4 master port used by the accelerator to write its results directly into system memory (SRAM/DDR3) at the address given by RESULT_BASE_ADDR. It performs two 64-bit write bursts per test (the score/index pair for the primary candidate, then the alternative), and asserts write_done once both writes are acknowledged by the bus — this is the signal STICKY_DONE latches on.

This accelerator does not use the AXI read channel (AR/R) — inputs are written directly into the register file by the CPU rather than DMA-read from memory, so those signals are intentionally left unused.

iv) top2_comparator.v

The scoring and selection core. Multiplies each of the 5 confidence/weight pairs (5 × 16-bit multipliers, mapped by Vivado to 5 DSP48E1 blocks) to compute a score per candidate, applies the score threshold filter to reject weak candidates, then compares all 5 scores to select the highest and second-highest scoring candidates as the primary and alternative outputs.

IV. FIRMWARE – MAIN.C

Location: ***Application/dummy_firmware_accelarator_64_64_matrix_20000/main.c***

The firmware initialises UART, then runs five predefined test cases back-to-back. Each test:

- Writes the 5 confidence/weight pairs to the accelerator's registers.
- Sets RESULT_BASE_ADDR to a unique memory slot for that test (0x30000 + test_index × 0x40), so all five results land at distinct, easy-to-find addresses.
- Flushes the data cache and pulses START.
- Polls STICKY_DONE (not the legacy READY bit) until the accelerator confirms results have actually been written to memory.
- Reads back RESULT_0 / RESULT_1 / VALID_MASK and writes a compact summary word to memory alongside the hardware's own write-back, as an independent software-side confirmation.

After all five tests, a fixed marker **`0xDEADC0DE`** — with is written to memory so the waveform can confirm the CPU reached the end of the program without hanging.

#	Task	Expected Primary	Expected Alternative
0	Water Plant	Potted Plant — 729900	Vase — 233600
1	Step On Something	Chair (1) — 514500	Chair (2) — 473900
2	Dig A Hole	Spoon — 630000	Fork — 450000
3	Sit Comfortably	Couch — 595000	Chair — 540000
4	Extinguish Fire	None — 0 (all filtered)	None — 0 (all filtered)

*Note: Certain tasks lacked an exact dataset match (e.g., Dig A Hole, with no native "shovel" class). For these cases, a **semantic fallback mechanism** selects the closest functional equivalent — e.g., **Spoon** and **Fork** for digging — ensuring a valid object is still returned.*

These five cases were chosen to exercise every distinct path through the scoring/filtering logic: a clear high-priority winner, a fallback to medium-priority objects, a task with no high-priority object defined at all, an independent second data point, and the edge case where every candidate falls below the score threshold.

Build

The firmware is compiled using the organiser-provided toolchain and script:

```
cd Application/ ./sim_app_compile_and_mif.sh
```

This produces accelerator_test.mif, which is loaded into DVCon_SoC_SRC/MEMORY_IP/rom_32KB_axi/ before simulation. The pre-built .mif is already included in this submission, so no rebuild is required to reproduce the results below.

V. WAVEFORMS

The design was simulated using **AMD Vivado Simulator (XSim)** against the organiser-provided, unmodified testbench. The following captures were taken from the same simulation run and show the accelerator correctly executing the test suite.



Figure 1: PROCESSOR BUS SIGNALS

VEGA processor AXI bus activity (proc_m0/proc_m1) during the test run. Read/write handshakes on the processor's master ports confirm the CPU is actively driving the bus while the accelerator's five tests execute in sequence.

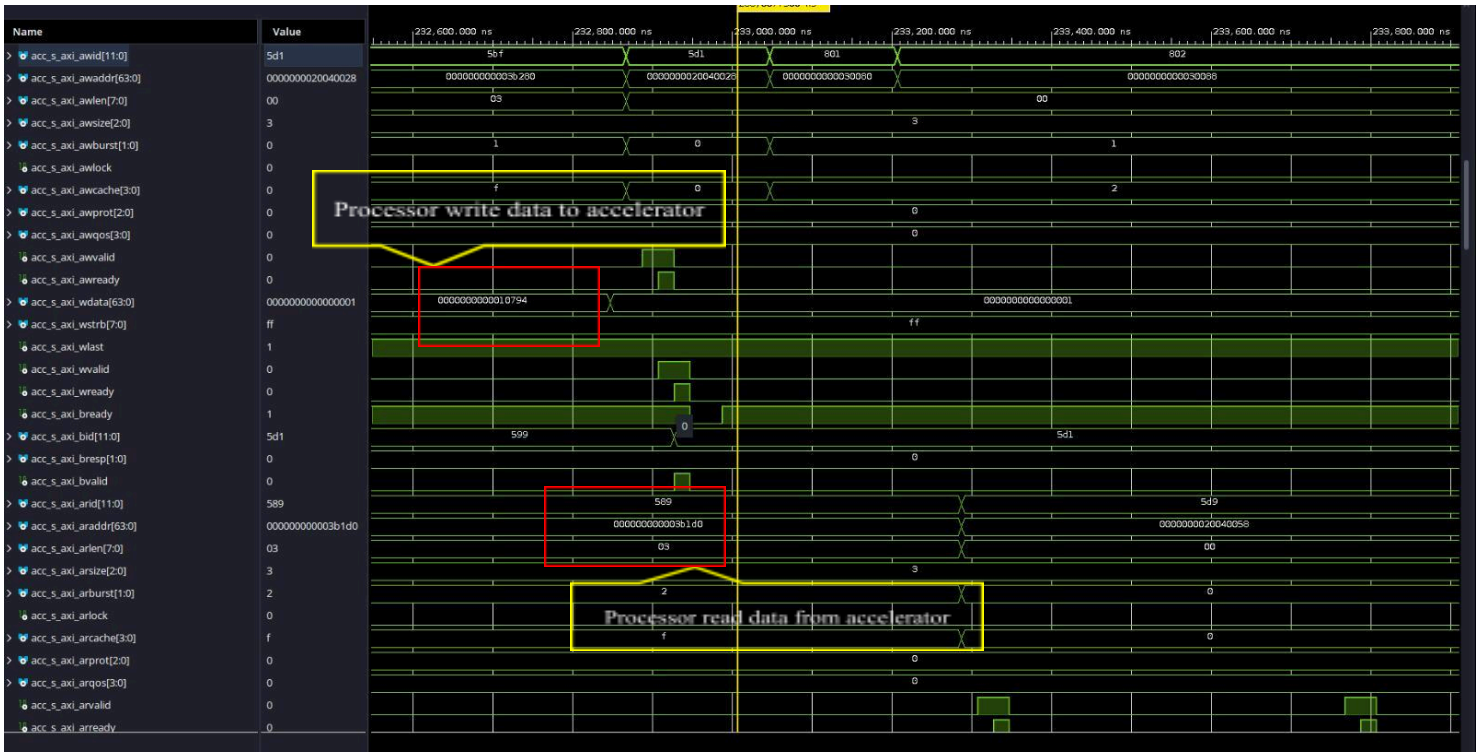


Figure 2: ACCELERATOR SLAVE-SIDE INTERFACE

CPU writes to the accelerator's register file over the AXI4 slave port. The write to address 0x20040028 corresponds to the START register (offset 0x28), confirming the processor correctly issues the start pulse for each test before polling for completion

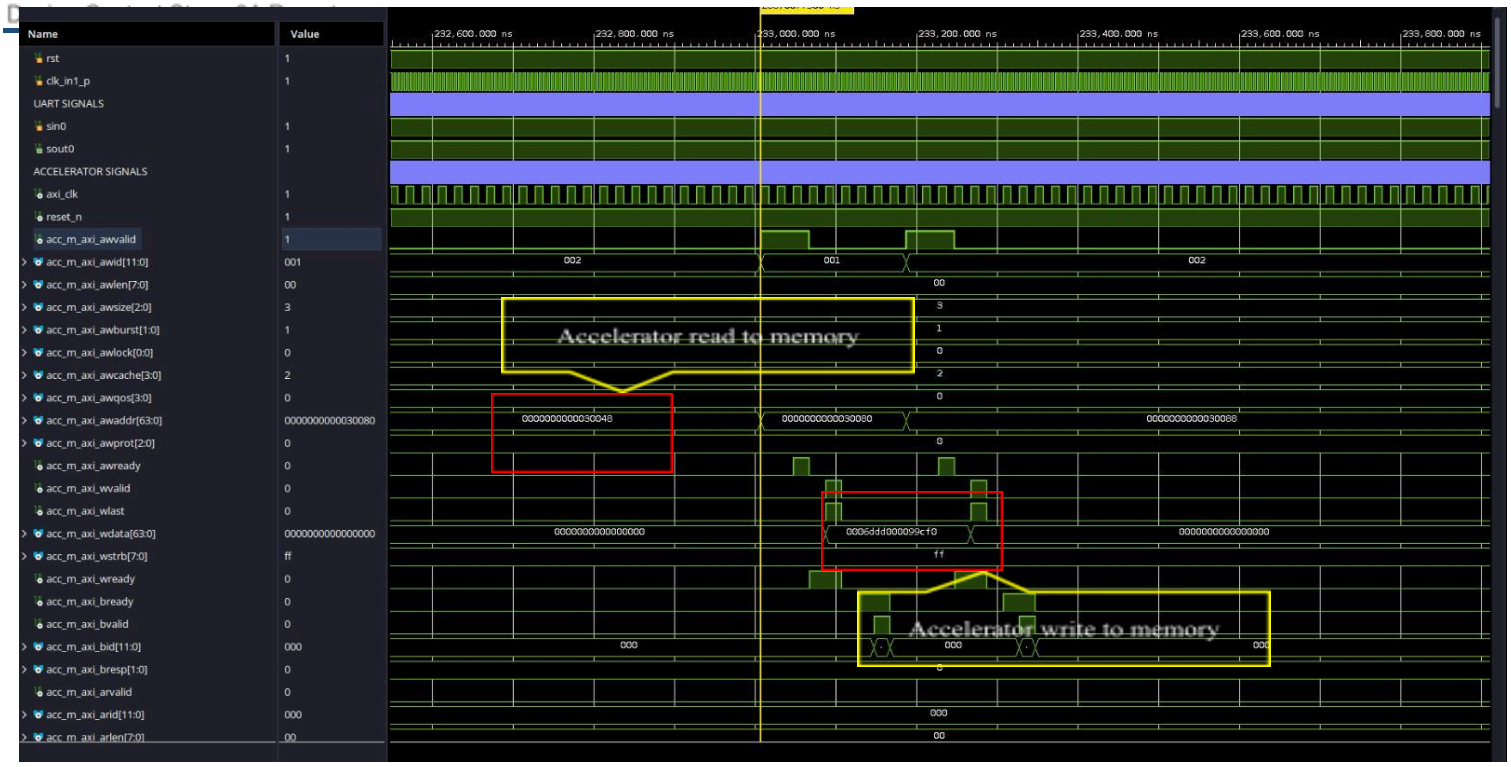


Figure 3: ACCELERATOR MASTER-SIDE INTERFACE

— Accelerator's AXI4 master write-back to memory. Write addresses step through 0x30048 → 0x30080 → 0x30088 as tests complete; write data 0x0006ddd000099cf0 splits into primary score 0x00099cf0 (630000) and alternative score 0x0006ddd0 (450000) — an exact match to the expected Test 2 ("Dig A Hole"



Figure 4: INTERNAL SCORING SIGNALS

Internal confidence inputs (confidence_0..4) and the five computed per-candidate scores (score0..score4) inside top2_comparator, alongside the final primary_score and alternative_score outputs. This confirms the weighted-score computation itself, not just the final AXI write-back, is functioning correctly for each test case.

VI . SYNTHESIS REPORTS

Synthesis was performed in AMD Vivado 2026.1, targeting the Digilent Genesys 2 board (xc7k325t-ffg900), as required by the contest's permitted-tools list. Full reports are included in GEN_BIT_OUT/;

Key results are summarised below.

a.Resource Utilization

Full-SoC utilisation (includes the baseline VEGA CPU, DDR3 controller, Ethernet, UART, etc., plus our accelerator):

Resource	Used	Available	Utilization
Slice LUTs	83,268	203,800	40.86%
Slice Registers	38,155	407,600	9.36%
Block RAM Tiles	81	445	18.20%
DSP48E1	55	840	6.55%

Our accelerator's own contribution (from the hierarchical utilisation report), isolated from the baseline SoC:

Module	LUTs	Registers	DSPs
u_Accelerator_Top (total)	841 (0.41%)	812 (0.20%)	5 (0.60%)
└ u_axi_master	39 (0.02%)	137 (0.03%)	0
└ u_axi_slave	199 (0.10%)	431 (0.11%)	0
└ top2_comparator (remainder)	603	244	5

The accelerator adds well under 1% to every resource category. The bulk of overall chip utilisation (~37% of the 40.86% LUT figure) is the pre-existing baseline VEGA CPU core, not our design.

b.Timing

Metric	Value	Result
WNS (Worst Negative Slack)	4.060 ns	Setup timing MET
TNS (Total Negative Slack)	0.000 ns	0 failing setup endpoints
WHS (Worst Hold Slack)	-4.246 ns	(pre-route)
THS Failing Endpoints	4,286 / 110,053	Expected at post-synthesis stage

Setup timing (WNS) is positive and clean, meeting the design's timing requirement at the synthesised (pre-route) stage. Hold-time violations (WHS) are expected and not meaningful at this stage — hold timing depends on real physical wire delays, which only exist after Place & Route.

c.Power

Metric	Value
Total On-Chip Power	0.564 W
Dynamic Power	0.394 W
Static (Device) Power	0.169 W
Junction Temperature	26.0 °C (well under the device maximum)
u_Accelerator_Top Power Draw	0.003 W (≈ 0.5% of total)
Confidence Level	Medium (expected pre-route)

The accelerator's power contribution is negligible relative to the rest of the SoC, consistent with its small resource footprint.

VII . CONCLUSION

The accelerator has been integrated into the VEGA SoC, verified in simulation across five representative test cases covering all major logic paths (clear winner, fallback tier, no-high-tier-exists, and all-filtered edge case), and synthesised cleanly with setup timing met and negligible resource/power overhead. All reports and waveform evidence are included alongside this document in ***GEN_BIT_OUT/*** and ***RESULTS_PROOF/***.